# CHAPTER 3

## SOCIAL NETWORKS

*A discussion of social networks and the empirical techniques used to probe their structure*

**S**OCIAL networks are networks in which the vertices are people, or sometimes groups of people, and the edges represent some form of social interaction between them, such as friendship. Sociologists have developed their own language for discussing networks: they refer to the vertices, the people, as *actors* and the edges as *ties*. We will sometimes use these words when discussing social networks.

We begin this chapter with a short summary of the origins and research focus of the field of social networks, before describing in detail some of the techniques used to discover social network structure. The material in this chapter forms the basis for understanding many of the social network examples that appear in the rest of the book.

## 3.1 THE EMPIRICAL STUDY OF SOCIAL NETWORKS

To most people the words "social network," if they mean anything, refer to online social networking services such as *Facebook* and *MySpace*. The study of social networks, however, goes back far farther than the networks' modernday computer incarnations. Indeed, among researchers who study networks, sociologists have perhaps the longest and best established tradition of quantitative, empirical work. There are clear antecedents of social network analysis to be found in the literature as far back as the end of the nineteenth century. The true foundation of the field, however, is usually attributed to psychiatrist Jacob Moreno, a Romanian immigrant to America who in the 1930s became interested in the dynamics of social interactions within groups of people. At a medical conference in New York City in March 1933 he presented the results of a set of studies he had performed that may have been the first true social network studies, and the work attracted enough attention to merit a column in the *New York Times* a few days later. A year after that Moreno published a book entitled *Who Shall Survive?* [228] which, though not a rigorous work by modern standards, contained the seeds of the field of *sociometry*, which later became social network analysis.

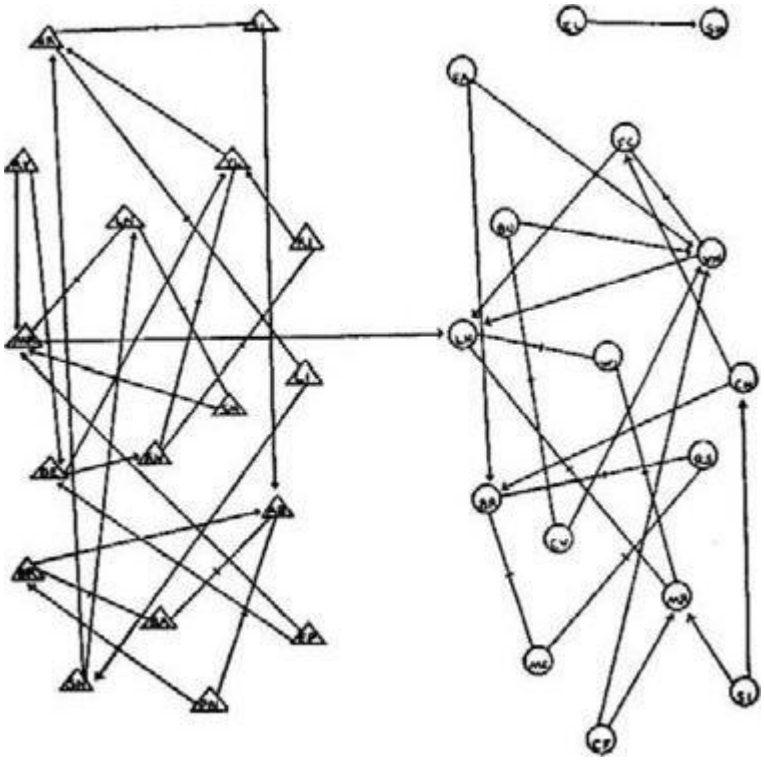


**Figure 3.1: Friendships between schoolchildren.** This early hand-drawn image of a social network, taken from the work of psychiatrist Jacob Moreno, depicts friendship patterns between the boys (triangles) and girls (circles) in a class of schoolchildren in the 1930s. Reproduced from [228] by kind permission of the American Society of Group Psychotherapy and Psychodrama.

Moreno called his diagrams of human interaction *sociograms*, rather than social networks (a term not coined until about twenty years later), but in everything but name they are clearly what we now know as networks. Figure 3.1, for instance, shows a hand-drawn figure from Moreno's book, depicting friendships within a group of schoolchildren. The triangles and circles represent boys and girls respectively and the figure reveals, among other things, that there are many friendships between two boys or two girls, but few between a boy and a girl. It is simple conclusions like this, that are both sociologically interesting and easy to see once one draws a picture of the network, that rapidly persuaded social scientists that there was merit in Moreno's methods.

One of the most important things to appreciate about social networks is that there are many different possible definitions of an edge in such a network and the particular definition one uses will depend on what questions one is interested in answering. Edges might represent friendship between individuals, but they could also represent professional relationships, exchange of goods or money, communication patterns, romantic or sexual relationships, or many other types of connection. If one is interested, say, in professional interactions between the boards of directors of Fortune 500 companies, then a network of who is dating whom or who looks at who else's Facebook page is probably not of much use. Moreover, the techniques one uses to probe different types of social interaction can also be quite different, so that different kinds of social network studies are typically needed to address different kinds of questions.

Direct questioning of experimental subjects is probably the most common method of determining the structure of social networks. We discuss it in detail in Section 3.2. Another important technique, the use of archival records (Sections 3.4 and 3.5), is illustrated by a different early example of a social network study. It was, apparently, a common practice in the US in the 1930s for newspapers to report on the public appearances of society women, and Davis, Gardner, and Gardner made use of this in a study of a social network of 18 women in a city in the American south. This study, often referred to in the literature as the "Southern Women Study," was described in a book by the researchers published in 1941 [86], although it was based on data from 1939. They took a sample of 14 social events attended by the women in question and recorded which women attended which events. Women in this network may be considered connected if they attended a common event. An alternative and more complete representation of the data is as an "affiliation network" or "bipartite graph," a network with two types of vertex, representing the women and the events, with edges connecting each woman to the events she attended. A visualization of the affiliation network for the Southern Women Study is shown in Fig. 3.2. One reason why this study has become so well known, in addition to its antiquity, is that the women were found by the researchers to split into two subgroups, tightly knit clusters of acquaintances with only rather loose between-cluster interaction. A classic problem in social network analysis is to devise a method or algorithm that can discover and extract such clustering from raw network data, and quite a number of researchers have made use of the Southern Women data as a test case for the development of such methods. Affiliation networks receive further attention in Section 3.5.

Such is the power of social network analysis that its techniques have, since Moreno and Davis *et al.*, been applied to an extraordinary variety of different communities, issues, and problems, including friendship and acquaintance patterns in local communities and in the population at large [36, 37, 175, 219, 311], and among students [334] and schoolchildren [112, 225, 277], contacts between business people and other professionals [78, 134], boards of directors of companies [87, 88, 207], collaborations of scientists [145, 146, 236], movie actors [16, 323], and musicians [139], sexual contact networks [183, 198, 272, 285] and dating patterns [34], covert and criminal networks such as networks of drug users [289] or terrorists [191], historical networks [259], online communities such as Usenet [204, 300, 312] or Facebook [196], and social networks of animals [205, 286, 287].

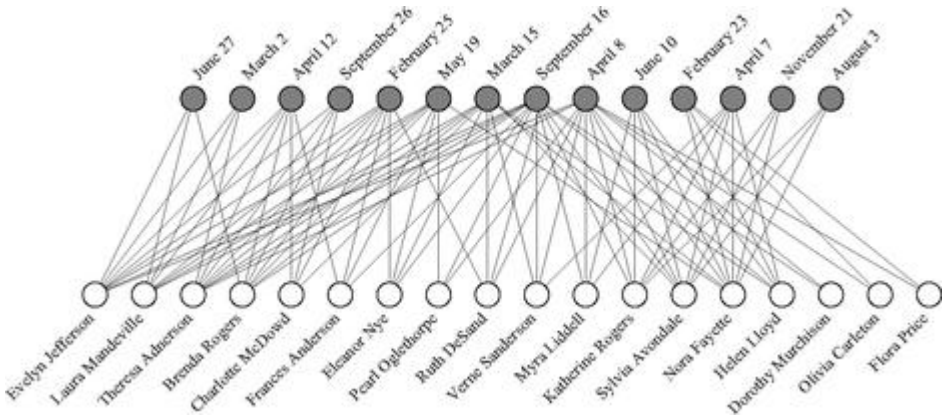



**Figure 3.2: The affiliation network of the "Southern Women Study."** This network (like all affiliation networks) has two types of vertex, the open circles at the bottom representing the 18 women who were the subjects of the study and the shaded circles at the top representing the social events they attended. The edges connect each woman to the events she attended, as deduced from newspaper reports. Data courtesy of L. Freeman and originally from Davis *et al.* [86].

We will see some examples of these and other networks throughout this book and we will give details as needed as we go along. The rest of the present chapter is devoted to a discussion of the different empirical methods used to measure social networks. The two techniques described above, namely direct questioning of subjects and the use of archival records, are two of the most important, but there are several others that find regular use. This chapter does not give a complete review of the subject—for that we refer the reader to specialized texts such as those of Wasserman and Faust [320] and Scott [293]—but we introduce as much material as will be needed for the later chapters of the book, while at the same time, we hope, giving some flavor for the challenges of empirical study in the field of social networks.

## 3.2 INTERVIEWS AND QUESTIONNAIRES

The most common general method for accumulating data on social networks is simply to ask people questions. If you are interested in friendship networks, then you ask people who their friends are. If you are interested in business relationships you ask people who they do business with, and so forth. The asking may take the form of direct interviews with participants or the completion by participants of questionnaires, either on paper or electronically. Indeed many modern studies, particularly surveys conducted by telephone, employ a combination of both interviews and questionnaires, wherein a professional interviewer reads questions from a questionnaire to a participant. By using a questionnaire, the designers of the study can guarantee that questions are asked, to a good approximation, in a consistent order and with consistent wording. By employing an interviewer to do the asking the study gains flexibility and reliability: interviewees often take studies more seriously when answering questions put to them by a human being, and interviewers may be given some latitude to probe interviewees when they are unclear, unresponsive, or confused. These are important considerations, since misunderstanding and inconsistent interpretation of survey questions are substantial sources of error. By making questions as uniform as possible and giving respondents personal help in understanding them, these errors can be reduced. A good introduction to social survey design and implementation has been given by Rea and Parker [279].

To find out about social networks, surveys typically employ a *name generator* , an item or series of items that invite respondents to name others with whom they have contact of a specified kind. For example, in their classic study of friendship networks among schoolchildren, Rapoport and Horvath [277] asked children to complete a questionnaire that included items worded as follows:

My best friend at ____ Junior High School is:
My second-best friend at ____ Junior High School is:
My third-best friend at ____ Junior High School is:
⋮
My eighth-best friend at ____ Junior High School is:

The blanks ″___″ in the questionnaire were filled in with the appropriate school name. The list stopped at the eighth-best friend and many children did not complete all eight.

Ideally all students within the school would be surveyed, though Rapoport and Horvath reported that in their case a few were absent on the day the survey was conducted. Note that the survey specifically asks children to name only friends within the school. The resulting network will therefore record friendship ties within the school but none to individuals outside. Since all social network studies are limited to some community or portion of the population, and since it is highly unlikely that such a community will have ties solely within the community and none outside, all surveys must make some decision about how to deal with ties to outside individuals. Sometimes they are recorded. Sometimes, as here, they are not. Such details can be important since statistics derived from the survey results will often depend on the decisions made.

There are some points to notice about the data produced by name generators. First, the network ties, friendships in the case above, are determined by one respondent nominating another by name. This is a fundamentally asymmetric process. Individual A identifies individual B as their friend. In many cases B will also identify A as *their* friend, but there is no guarantee that this will happen

and it is not uncommon for nomination to go only one way. We normally think of friendship as a two-way type of relationship, but surveys suggest that this not always the case. As a result, data derived from name generators are often best represented as directed networks, networks in which edges run in a particular direction from one vertex to another. If two individuals nominate each other then we have two directed edges, one pointing in either direction. Each vertex in the network also has two degrees, an outdegree—the number of friends identified by the corresponding individual—and an in-degree—the number of others who identified the individual as a friend.

> We encountered directed networks previously in Chapter 1, in our discussion of the World Wide Web, and they are discussed in more detail in Section 6.4.

This brings us to a second point about name generators. It is common, as in the example above, for the experimenter to place a limit on the number of names a respondent can give. In the study of Rapoport and Horvath, this limit was eight. Studies that impose such a limit are called *fixed choice* studies. The alternative is to impose no limit. Studies that do this are called *free choice* studies.

Limits are often imposed purely for practical purposes, to reduce the work the experimenter must do. However, they may also help respondents understand what is required of them. In surveys of schoolchildren, for instance, there are some children who, when asked to name all their friends, will patiently name all the other children in the entire school, even if there are hundreds of them. Such responses are not particularly helpful in surveys—almost certainly the children in question are employing a definition of friendship different from that employed by most of their peers and by the investigators.

However, limiting the number of responses is for most purposes undesirable. In particular, it clearly limits the out-degree of the vertices in the network, imposing an artificial and possibly unrealistic cut-off. As discussed in Chapter 1, an interesting property of many networks is the existence of a small number of vertices with unusually high degree, and it is known that in some cases these vertices, though few in number, can have a dominant effect on the behavior of the network as a whole. By employing a name generator that artificially cuts off the degree, any information about the existence of such vertices is lost.

It is worth noticing, however, that even in a fixed-choice study there is normally no limit on the *in*-degree of vertices in the network; there is no limit to the number of times an individual can be nominated by others. And indeed in many networks it is found that a small number of individuals are nominated an unusually large number of times. Rapoport and Horvath [277] observed this in their friendship networks: while most children in a school are nominated as a friend of only a few others, a small number of popular children are nominated very many times. Rapoport and Horvath were some of the first scientists in any field to study quantitatively the degree distribution of a network, reporting and commenting extensively on the in-degrees in their friendship networks.

Not all surveys employing name generators produce directed networks. Sometimes we are interested in ties that are intrinsically symmetric between the two parties involved, in which case the edges in the network are properly represented as undirected. An example is networks of sexual contact, which are widely studied to help us understand the spread of sexually transmitted diseases [183, 198, 272, 285]. In such networks a tie between individuals A and B means that A and B had sex. While participants in studies sometimes do not remember who they had sex with or may be unwilling to talk about it, it is at least in principal a straightforward yes-or-no question whether two people had sex, and the answer should not depend on which of the two you ask.[12] In such networks therefore, ties are normally represented as undirected.

Surveys can and often do ask respondents not just to name those with whom they have ties but to describe the nature of those ties as well. For instance, questions may ask respondents to name people they both like and dislike, or to name those with whom they have certain types of contact, such as socializing together, working together, or asking for advice. For example, in a study of the social network of a group of medical doctors, Coleman *et al.* [78] asked respondents the following questions:

> Who among your colleagues do you turn to most often for advice?

With whom do you most often discuss your cases in the course of an ordinary week?

Who are the friends among your colleagues who you see most often socially?

The names of a maximum of three doctors could be given in response to each question. A survey such as this, which asks about several types of interactions, effectively generates data on several different networks at once—the network of advice, the discussion network, and so forth.

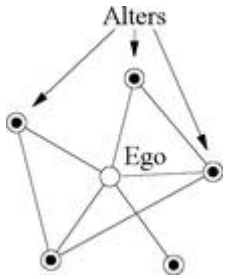
Surveys may also pose questions aimed at measuring the strength of ties, asking for instance how often people interact or for how long, and they may ask individuals to give a basic description of themselves: their age, income, education, and so forth. Some of the most interesting results of social network studies concern the extent to which people's choice of whom they associate with reflects their own background and that of their associates. For instance, you might choose to socialize primarily with others of a similar age to yourself, but turn for advice to those who are older than you.

The main disadvantages of network studies based on direct questioning of participants are that they are first laborious and second inaccurate. The administering of interviews or questionnaires and the collation of responses is a demanding job that has been only somewhat helped in recent years by the increasing availability of computers and the use of online survey tools. Most studies have been limited to a few tens or at most hundreds of respondents—the 34-vertex social network of Fig. 1.2 is a typical example. It is a rare study that contains more than a thousand actors, and studies such as the National Longitudinal Study of Adolescent Health,[13] which compiled responses from over 90 000 participants, are very unusual and extraordinarily costly. Only a substantial public interest such as, in that case, the control of disease, can justify their funding.

Data based on direct questioning are also plagued by uncontrolled biases. Answers given by respondents are always, to some extent, subjective. If you ask people who their friends are, different people will interpret "friend" in different ways and thus give different kinds of answers. Investigators do their best to pose questions and record answers in a uniform fashion, but it is inevitable that inconsistencies will be present in the final data and anyone who has ever conducted a survey knows this well. This problem is not unique to social network studies. Virtually all social surveys suffer from such problems and a large body of expertise has been developed concerning techniques for dealing with them. Nonetheless, one should bear in mind when dealing with any social network data derived from interviews or questionnaires the possibility of uncontrolled experimental bias in the results.

### 3.2.1 EGO-CENTERED NETWORKS

Studies of the type described in the previous section, in which all or nearly all of the individuals in a community are surveyed, are called *sociometric* studies, a term coined by Jacob Moreno himself (see the discussion at the beginning of this chapter). For the purposes of determining network structure, sociometric studies are desirable; unless we survey all or nearly all of the population of interest, there is no way we can reconstruct the complete network of ties within that population. However, as discussed at the end of the preceding section, sociometric studies also require a lot of work and for large populations may simply be infeasible.




An ego-centered network consisting of an ego and five alters.

At the other end of the spectrum lie studies of *personal networks* or *ego-centered networks*.[14] An ego-centered network is the network surrounding one particular individual, meaning, usually, the individual surveyed and his or her immediate contacts. The individual surveyed is referred to as the *ego* and the contacts as *alters*.

The typical survey of this kind is conducted using direct questioning techniques similar to those discussed in Section 3.2, with interviews, questionnaires, or a combination of both being the instruments of choice. One might, for instance, select a sample of the target population at random,[15] and ask them to identify all those with whom they have a certain type of contact. Participants might also be asked to describe some characteristics both of themselves and of their alters, and perhaps to answer some other simple questions, such as which alters also have contact with one another.

Obviously surveys of this type, and studies of ego-centered networks in general, cannot reveal the structure of an entire network. One receives snapshots of small local regions of the network, but in general those regions will not join together to form a complete social network. There are cases, however, where we are primarily interested in local network properties, and egocentered network studies can give us good data about these. For example, if we wish to know about the degrees of vertices in a network then a study in which a random sample of people are each asked to list their contacts can give us reasonable degree statistics. (Studies probing vertex degrees are discussed more below.) If we also gather data on the contacts between alters, we can estimate clustering coefficients (see Section 7.9). If we have data on characteristics of egos and alters we can measure assortative mixing (Sections 7.13 and 8.7).

An example of a study gathering ego-centered network data is the General Social Survey (GSS) [59], a large-scale survey conducted every year in the United States since 1972 (every two years since 1994). The GSS is not primarily a social network study. The purpose of the study is to gather data about life in the United States, how it is changing, and how it differs from or relates to life in

other societies. The study contains a large number of items ranging from general questions probing the demographics and attitudes of the participants, to specific questions about recent events, political topics, or quality of life. However, among these many items there are in each iteration of the survey a few questions about social networks. The precise number and wording of these questions changes from one year to another, but here some examples from the survey of 1998, which was fairly typical:

> From time to time, most people discuss important matters with other people. Looking back over the last six months, who are the people with whom you discussed matters important to you? Do you feel equally close to all these people?
>
> Thinking now of close friends—not your husband or wife or partner or family members, but people you feel fairly close to—how many close friends would you say you have? How many of these close friends are people you work with now? How many of these close friends are your neighbors now?

And so on. By their nature these questions are of a "free choice" type, the number of friends or acquaintances the respondent can name being unlimited, although (and this is a criticism that has been leveled at the survey) they are also quite vague in their definitions of friends and acquaintances, so people may give answers of widely varying kinds.

Another example of an ego-centered network study is the study by Bernard *et al.* [36, 37, 175, 213] of the degree of individuals in acquaintance networks (i.e., the number of people that people know). It is quite difficult to estimate how many people a person knows because most people cannot recall at will all those with whom they are acquainted and there is besides a large amount of variation in people's subjective definition of "knowing." Bernard *et al.* came up with an elegant experimental technique to circumvent these difficulties. They asked people to read through a list containing a sample of several hundred family names drawn from a telephone directory.[16] Participants counted up how many people they knew with names appearing on the list. Each person with a listed name was counted separately, so that two acquaintances called "Smith" would count as two people. They were instructed to use the following precise definition of acquaintance:

> You know the person and they know you by sight or by name; you can contact them in person by telephone or by mail; and you have had contact with the person in the past two years.

(Of course, many other definitions are possible. By varying the definition, one could probe different social networks.) Bernard *et al.* then fed the counts reported by participants into a statistical formula to estimate the total number of acquaintances of each participant.

Bernard *et al.* repeated their study with populations drawn from several different cities and the results varied somewhat from city to city, but overall they found that the typical number of acquaintances, in the sense defined above, of the average person in the United States is on the order of about 2000. In the city of Jacksonville, Florida, for instance, they found a figure of 1700, while in Orange County, California they found a figure of 2025. Many people find these numbers surprisingly high upon first encountering them, perhaps precisely because we are poor at recalling all of the many people we know. But repeated studies have confirmed figures of the same order of magnitude, at least in the United States. In some other countries the figures are lower. In Mexico City, for instance, Bernard *et al.* estimated that the average person knows about 570 others.

## 3.3 DIRECT OBSERVATION

An obvious method for constructing social networks is direct observation. Simply by watching interactions between individuals one can, over a period of time, form a picture of the networks of unseen ties that exist between those individuals. Most of us, for instance, will be at least somewhat aware of friendships or enmities that exist between our friends or coworkers. In direct observation studies, researchers attempt to develop similar insights about the members of the population of interest.

Direct observation tends to be a rather labor-intensive method of study, so its use is usually restricted to rather small groups, primarily ones with extensive face-to-face interactions in public settings. In Chapter 1 we saw one such example, the "karate club" network of Zachary [334]. Another example is the study by Freeman *et al.* [131, 132] of the social interactions of windsurfers on a beach. The experimenters simply watched the individuals in question and recorded the length in minutes of every pairwise interaction among them. A large number of direct-observation network data sets were compiled by Bernard and co-workers during the 1970s and 80s as part of a lengthy study of the accuracy of individuals' perception of their own social situation [38, 40, 41, 173]. These include data sets on interactions between students, faculty, and staff in a university department, on members of a university fraternity,[17] on users of a teletype service for the deaf, and several other examples.

One arena in which direct observation is essentially the only viable experimental technique is in studies of the social networks of animals—clearly animals cannot be surveyed using interviews or questionnaires. One method is to record instances of animal pairs engaging in recognizable social behaviors such as mutual grooming, courting, or close association and then to declare ties to exist between the pairs that engage in these behaviors most often [205]. Not all animal species form interesting or useful social networks, but informative studies have been performed of, amongst others, monkeys [121, 286, 287], kangaroos [143], and dolphins [80, 205]. Networks in which the ties represent aggressive behaviors have also been reported, such as networks of baboons [214], wolves [163, 316], and ants [77]. In cases where aggressive behaviors normally result in one animal's establishing dominance over another the resulting networks can be regarded as directed and are sometimes called *dominance hierarchies* [90, 91, 101].

## 3.4 DATA FROM ARCHIVAL OR THIRD-PARTY RECORDS

An increasingly important, voluminous, and often highly reliable source of social network data is archival records. Such records are, sometimes at least, relatively free from the vagaries of human memory and are often impressive in their scale, allowing us to construct networks of a size that would require far more effort were other techniques used.

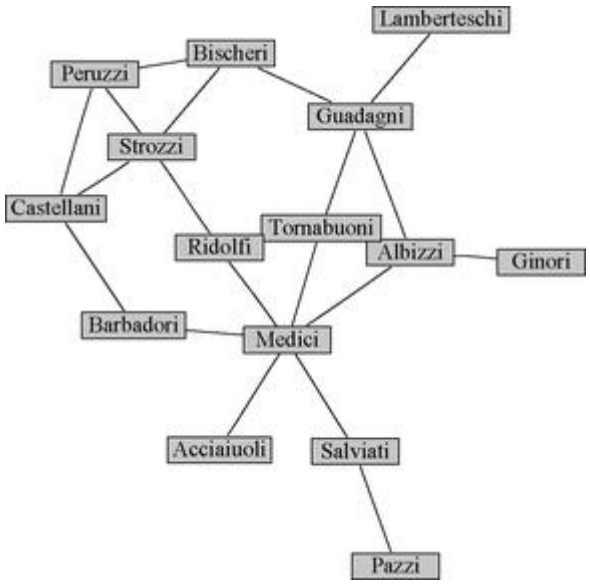



**Figure 3.3: Intermarriage network of the ruling families of Florence.** In this network the vertices represent fifteenth century Florentine families and the edges represent ties of marriage between them. After Padgett and Ansell [259].

A well-known small example of a study based on archival records is the study by Padgett and Ansell of the ruling families of Florence in the fifteenth century [259]. In this work, the investigators looked at contemporaneous historical records to determine which among the families had trade relations, marriage ties, or other forms of social contact with one another. Figure 3.3 shows one of the resulting networks, a network of intermarriages between 15 of the families. It is notable that the Medici family occupies a central position in this network, having marriage ties with members of no fewer than six other families. Padgett and Ansell conjectured that it was by shrewd manipulation of social ties such as these that the Medici rose to a position of dominance in Florentine society.

In recent years, with the widespread availability of computers and online databases, many more networks have been constructed from records of various types. A number of authors, for example, have looked at email networks [103, 313]. Drawing on email logs—automatic records kept by email servers of messages sent—it is possible to construct networks in which the vertices are people (or more correctly email addresses) and the directed edges between them are email messages. Exchange of email in such a network can be taken as a proxy for acquaintance between individuals, or we may be interested in the patterns of email exchange for some other reason. For

instance, email messages can carry computer viruses and a knowledge of the structure of the network of messages may help us to predict and control the spread of those viruses.

Another form of email network is the network formed by email address books. An email address book is a computer file in which a computer user stores, for convenience, the email addresses of his or her regular correspondents. The set of all such address books can be regarded as defining a network in which the vertices represent the owners of the address books, and there is a directed edge from vertex A to vertex B if person B's address appears in person A's address book. This network is again of interest in the study of computer viruses, since some viruses search address books for the addresses of new victims to infect and hence spread over the address book network. Similar networks can also be constructed for other forms of electronic communication that use address books, such as instant messaging [301].

> Telephone call graphs are quite distinct from the physical network of telephone cables discussed in Section 2.2. Indeed, a call graph is to the physical telephone network roughly as an email network is to the Internet.

A form of network similar to but older than the email network is the *telephone call graph*. In such a network the vertices represent telephone numbers and directed edges between them represent telephone calls from one number to another. Call graphs can be reconstructed from call logs kept by telephone companies, although such logs are generally proprietary and not easily available outside of those companies, and call graphs have as a result only occasionally been examined in the scientific literature [1, 9, 258].

Recent years have seen the rapid emergence of online social networking services, such as *Facebook* and *Linkedln*, which exist primarily to promote, document, and exploit the networks of contacts between individuals. As a natural part of their operation, these services build records of connections between their participants and hence provide, at least in principle, a rich source of archival network data. These data, however, like those for telephone calls, are largely proprietary to the companies operating the services and hence quite difficult to get hold of. So far only a few studies have been published of online social networks [53], but internal studies have no doubt been performed by the companies themselves and it is only a matter of time before more data become publicly available.

A few other online communities, not explicitly oriented towards networks, have been studied using network techniques. For instance, Holme *et al.* [158] took records of interactions between members of a Swedish dating website and reconstructed from them the network of interactions between the site's members. This study was unusual in that the network was time-resolved—the date and time of each interaction were recorded, allowing investigators to reconstruct after the fact the timing and duration of contacts between individuals. Most of the sources of network data considered in this book are not time-resolved, but many of the networks they correspond to do nonetheless change over time. Time-resolved network studies, or *longitudinal* studies, as they are called in sociology, are certainly a growth area to watch for in the future.

Another source of network data representing online communities is the Internet newsgroup system *Usenet*, a worldwide online message-board system that allows users to post messages on a large variety of topics. Messages are date and time stamped and identified with the name or email address of the poster along with a unique reference number that allows a poster to indicate when a posting is a reply or follow-on to a previous posting. Thus one can reconstruct the thread of the conversation taking place in a newsgroup, and in particular assemble a network in which the vertices are posters and the edges represent a response by one poster to a posting by another. Studies of newsgroup networks of this kind have been performed by a number of authors [204, 300, 312].

Weblogs and online journals are another source of online social network data. Online journals of various kinds have become popular on the World Wide Web since around the turn of the century. On these websites the proprietor posts whatever thoughts he or she cares to make public, along with links to sites maintained by others. These links form a directed network that lies, in terms of semantic content, somewhere between a social network and the World Wide Web; the links are often informational—the linker wishes to bring to his or her readers' attention the contents of the

linked site—but there is a strong social element as well, since people often link to sites operated by their friends or acquaintances. This trend is particularly noticeable within journal communities such as *LiveJournal* and among weblogs devoted to specific topics, such as science or politics. The structure of the networks of links can be extracted using "crawlers" similar to those used to search the Web—see Section 4.1. Studies of journals and weblogs have been performed for example by Adamic and Glance [4] and MacKinnon and Warren [206].

An interesting network that has features of both a social and a technological network is the network of trust formed by a set of cryptographic keys. Cryptosystems or cyphers (i.e., secret codes), long associated in the public mind with spies and skulduggery, have become a crucial part of the twenty-first-century economy, used to protect important data, particularly financial data such as credit card numbers, from theft and misuse. An important advance, central to the widespread and convenient use of cryptography, was the development in the 1970s of *public-key cryptography*. In traditional cryptosystems, two parties wishing to exchange messages must share a *key* that they use to encode and decode the messages. The key is typically a large number, which is used in combination with the chosen cryptosystem to dictate exactly how messages are converted from their original "plain text" form into code and back again. This key, which allows anyone possessing it to decode the messages, must be kept secret from any malicious snoopers, and this raises the difficult problem of how the parties in question agree on the key in the first place. Usually the key is generated by a computer program run by one party, but then it must be transmitted securely to the other party without anyone else seeing it. Sending the key over the Internet unencrypted would pose a significant risk of detection. Physical transmission, for example by conventional mail, would be reasonably secure, but would take a long time. Most customers buying goods over the Internet would not want to wait a week for a secret key to arrive by mail from their vendor.

These problems were solved with the invention of public-key cryptography in the 1970s. Public-key cryptosystems make use of any of several different *asymmetric cyphers* in which two different keys are used. One key, called the *public key*, is employed in the computer algorithm that converts the message from plain text into its encrypted form, but a different key, the *private key*, is needed to decrypt the message. The public key cannot be used for decryption.[18] The two keys are generated as a pair by one of the two parties wishing to exchange information and the public key is sent openly over the Internet or other channel to the second party. The private key remains the secret property of the first party and is not shared. The second party can then send messages to the first by encoding them with the public key and only the first party can decode them.[19] Although the public key can be intercepted by a third party in transmission, it will do the third party no good, since the public key cannot be used to decode messages, only to encode them. Indeed, in many cases, users of public-key systems deliberately broadcast their public keys to anyone who might want them, inviting the world to send them encoded messages, messages which only they can decode. It is from such practices that the name "public-key cryptography" arises.

Some asymmetric cyphers can also be used in the reverse direction. That is, one can encode a message with the private key and it can only be decoded with the public key. Why would one want to do this, when everyone has the public key? The answer is that you can use it to prove your identity. Someone talking to you over the Internet, say, may want to be certain that you are who you claim to be (rather than some nefarious interloper) before they trust you with, for instance, their credit card number. So they send you a specific message that they choose, usually just a random string of digits, and ask you to encrypt it using your private key. Having done so, you send the encrypted message back to them and they decode it with the public key. If the decoded message matches the original one then they know that you are who you say you are, since no one else has your private key and hence no one else could have encrypted a message that decodes correctly with the public key.[20] This "digital signature" process is a crucial part of electronic commerce, allowing buyers and sellers to confirm each other's identities before doing business, and is used millions of times every day in transactions of every sort.

But there is still a fundamental problem with public-key encryption, namely the problem of knowing that the public key you are given really was created by the person you think it was created by. Some malicious person could create a public/private key pair and broadcast the public key to the world, labeled with, say, the name of a bank or retail trader, then use that key in a digital

signature scheme to persuade some unsuspecting victim that they are the trader and that the victim should send them a credit card number.

One way around this problem is to have people sign each other's public keys [267]. That is, party A takes a public key that claims to belong to party B, and that A knows in fact to be genuine, and encrypts it with their own private key. Now if you have A's public key and you believe it to be genuine, then you can take the encrypted key and decode it with A's public key, thereby recovering B's public key, which A says is genuine. If you trust A to make this statement, then you can now also trust that the key you have is B's true public key.

But now one can repeat the process. Now that you have a genuine public key for party B, and if you trust B, then B can now sign the keys that *they* know to be genuine and you will be able to verify that they are genuine also. In this way, parties who trust each other can securely represent to one another that keys are genuine.

The act of digitally signing someone else's public key is equivalent to saying that you know, or at least believe, the public key to be genuine, belonging to the person it claims to belong to. That act can be represented by a directed edge in a network. The vertices in the network represent the parties involved and a directed edge from party A to party B indicates that A has signed B's public key. The resulting directed network certainly has technological aspects but is in many ways more of a social network than anything else. People tend to vouch for the keys of other people they know, people they have communicated with or worked with frequently, so that they have both a good idea that the key in question is indeed genuine and a personal reason for making the effort to sign it.

Since public keys and the digital signatures of the people who sign them are, by necessity, public, it is relatively easy to construct a key-signing network from widely available data. There are a number of widely used key-signing networks associated, usually, with particular commercial cryptography products. One of the largest, for instance, is the network associated with the cryptography program PGP [267]. There have been only a small number of studies so far of the properties of key signing networks [47, 148] but there are certainly interesting questions awaiting answers in this area.

## 3.5 AFFILIATION NETWORKS

An important special case of the reconstruction of networks from archival records is the *affiliation network*. An affiliation network is a network in which actors are connected via comembership of groups of some kind. We saw one example in the introduction to this chapter, the Southern Women Study of Davis *et al.* [86], in which the authors drew their data from newspaper reports of social events and the "groups" were the sets of individuals who attended particular events. As we saw, the most complete representation of an affiliation network is as a network with two types of vertex representing the actors and the groups, with edges connecting actors to the groups to which they belong—see Fig. 3.2 on page 39. In such a representation, called a "bipartite network" or "two-mode network," there are no edges connecting actors directly to other actors (or groups to other groups), only actors to groups.

We study bipartite networks in more detail in Section 6.6.

Many examples of affiliation networks can be found in the literature. Another famous case is the study by Galaskiewicz [134] of the CEOs of companies in Chicago in the 1970s and their social interaction via clubs that they attended. In this network the CEOs are the actors and the clubs are the groups. Also in the business domain, quite a number of studies have been conducted of the boards of directors of companies [87, 88, 207]. In these networks the actors are company directors and the groups are the boards on which they sit. In addition to looking at the connections between directors in such networks, which arise as a result of their sitting on boards together, a considerable amount of attention has also been focused on the connections between boards (and hence between companies) that arise as a result of their sharing a common director, a so-called board "interlock."

More recently, some extremely large affiliation networks have been studied in the mathematics and physics literature. Perhaps the best known example is the network of collaborations of film actors, in which the "actors" in the network sense are actors in the dramatic sense also, and the groups to which they belong are the casts of films. This network is the basis, among other things, for a well-known parlor game, sometimes called the "Six Degrees of Kevin Bacon," in which one attempts to connect pairs of actors via chains of intermediate costars in a manner reminiscent of the small-world experiments of Stanley Milgram, which we discuss in Section 3.6. The film actor network has, with the advent of the Internet, become very thoroughly documented and has attracted the attention of many network analysts in recent years [16, 27, 323], although it is not clear whether there are any conclusions of real scientific interest to be drawn from its study.

Another example of a large affiliation network, one that holds more promise of providing useful results, is the coauthorship network of academics. In this network an actor is an academic author and a group is the set of authors of a learned paper. Like the film actor network, this network has become well documented in the last few years with the appearance of extensive online bibliographic resources covering many areas of human endeavor. Whether one is interested in papers published in journals or in more informal forums such as online preprint servers, excellent records now exist in most academic fields of authors and the papers they write, and a number of studies of the corresponding affiliation networks have been published [29, 89, 145, 146, 234-236].

## 3.6 THE SMALL-WORLD EXPERIMENT

An unusual contribution to the social networks literature was made by the experimental psychologist Stanley Milgram in the 1960s with his now-famous "small-world" experiments [219, 311]. Milgram was interested in quantifying the typical distance between actors in social networks. As discussed in Chapter 1, the "geodesic distance" between two vertices in a network is the minimum number of edges that must be traversed to travel from one vertex to the other through the network. Mathematical arguments suggest (as we will see later in this book) that this distance should be quite small for most pairs of vertices in most networks, a fact that was already well known in Milgram's time.[21] Milgram wanted to test this conjecture in real networks and to do this he concocted the following experiment.[22]

Milgram sent a set of packages, 96 in all, to recipients randomly chosen from the telephone directory in the US town of Omaha, Nebraska. The packages contained an official-looking booklet, or "passport," emblazoned with the crest of Milgram's home institution, Harvard University. Written instructions were included asking the recipients to attempt to get the passport to a specified target individual, a friend of Milgram's who lived in Boston, Massachusetts, over a thousand miles away. The only information supplied about the target was his name (and hence indirectly the fact that he was male), his address, and his occupation as a stockbroker. But the passport holders were not allowed simply to send their passport to the given address. Instead they were asked to pass it to someone they knew on a first-name basis and more specifically the person in this category who they felt would stand the best chance of getting the passport to the intended target. Thus they might decide to send it to someone they knew who lived in Massachusetts, or maybe someone who worked in the financial industry. The choice was up to them. Whoever they did send the passport to was then asked to repeat the process, sending it on to one of *their* acquaintances, so that after a succession of such steps the passport would, with luck, find its way into the hands of its intended recipient. Since every step of the process corresponded to the passport's changing hands between a pair of first-name acquaintances, the entire path taken corresponded to a path along the edges of the social network formed by the set of all such acquaintanceships. Thus the length of the path taken provided an upper bound on the geodesic distance in this network between the starting and ending individuals in the chain.

Of the 96 passports sent out, 18 found their way to the stockbroker target in Boston. While this may at first sound like a low figure, it is actually remarkably high—recent attempts to repeat Milgram's work have resulted in response rates orders of magnitude lower [93]. Milgram asked participants to record in the passport each step of the path taken, so he knew, among other things, how long each path was, and he found that the mean length of completed paths from Omaha to the target was just 5.9 steps. This result is the origin of the idea of the "six degrees of separation," the popular belief that there are only about six steps between any two people in the world.[23]

There are of course many reasons why this result is only approximate. Milgram used only a single target in Boston, and there is no guarantee the target was in any way typical of the population as a whole. And all the initial recipients in the study were in a single town in the same country.[24] (None of the completed chains that reached the target went outside the country.) Also there is no guarantee that chains took the shortest possible route to the target. Probably they did not, at least in some cases, so that the lengths of the paths found provide, as we have said, only an upper bound on the actual geodesic distance between vertices. And most of the chains of course were never completed. The passports were discarded or lost and never made their way to the target. It is reasonable to suppose that the chances of getting lost were greater for passports that took longer paths, and hence that the paths that were completed were a biased sample, having typical lengths shorter than the average.

For all these reasons and several others, Milgram's experiments should be taken with a large pinch of salt. Even so, the fundamental result that vertex pairs in social networks tend on average to be connected by short paths is now widely accepted, and has moreover been shown to extend to many other kinds of networks as well. Enough experiments have confirmed the effect in enough networks that, whatever misgivings we may have about Milgram's particular technique, the general result is not seriously called into question.

Funneling is discussed further in Section 8.2.

Milgram's experiments also, as a bonus, revealed some other interesting features of acquaintance networks. For instance, Milgram found that most of the passports that did find their way to the stockbroker target did so via just three of the target's friends. That is, a large fraction of the target's connections to the outside world seemed to be through only a few of his acquaintances, a phenomenon sometimes referred to as the "funneling" effect. Milgram called such well-connected acquaintances "sociometric superstars," and their existence has occasionally been noted in other networks also, such as collaboration networks [234], although not in some others [93].

A further interesting corollary of Milgram's experiment has been highlighted by Kleinberg [177, 178]. (Milgram himself seems not to have appreciated the point.) The fact that a moderate number of the passports did find their way to the intended target person shows not only that short paths exist in the acquaintance network, but also that people are good at finding those paths. Upon reflection this is quite a surprising result. As Kleinberg has shown, it is possible and indeed common for a network to possess short paths between vertices but for them to be hard to find unless one has complete information about the structure of the entire network, which the participants in Milgram's studies did not. Kleinberg has suggested a possible explanation for how participants found the paths they did, based on conjectures about the structure of the network. We discuss his ideas in detail in Section 19.3.

Recently the small-world experiment has been repeated by Dodds *et al.* [93] using the modern medium of email. In this version of the experiment participants forwarded email messages to acquaintances in an effort to get them ultimately to a specified target person about whom they were told a few basic facts. The experiment improved on that of Milgram in terms of sheer volume, and also by having much more numerous and diverse target individuals and starting points for messages: 24 000 chains were started, most (though not all) with unique starting individuals, and with 18 different participating targets in 13 different countries. On the other hand, the experiment experienced enormously lower rates of participation than Milgram's, perhaps because the public is by now quite jaded in its attitude towards unsolicited mail. Of the 24 000 chains, only 384, or 1.5%, reached their intended targets, compared with 19% in Milgram's case. Still, the basic results were similar to those of Milgram. Completed chains had an average length of just over four steps. Because of their better data and considerably more careful statistical analysis, Dodds *et al.* were also able to compensate for biases due to unfinished chains and estimated that the true average path length for the experiment was somewhere between five and seven steps—very similar to Milgram's result. However, Dodds *et al.* observed no equivalent of the "sociometric superstars" of Milgram's experiment, raising the question of whether their appearance in Milgram's case was merely a fluke of the particular target individual he chose rather than a generic property of social networks.

An interesting variant on the small-world experiment has been proposed by Killworth and Bernard [39, 174], who were interested in how people "navigate" through social networks, and specifically how participants in the small-world experiments decide whom to forward messages to in the effort to reach a specified target. They conducted what they called "reverse small-world" experiments[25] in which they asked participants to *imagine* that they were taking part in a small-world experiment. A (fictitious) message was to be communicated to a target individual and participants were asked what they wanted to know about the target in order to make a decision about whom to forward the message to. The actual passing of the message never took place; the experimenters merely recorded what questions participants asked about the target. They found that three characteristics were sought overwhelmingly more often than any others, namely the name of the target, their geographic location, and their occupation—the same three pieces of information

that Milgram provided in his original experiment. Some other characteristics came up with moderate frequency, particularly when the experiment was conducted in non-Western cultures or among minorities: in some cultures, for instance, parentage or religion were considered important identifying characteristics of the target.

> The mechanisms of network search and message passing are discussed in greater detail in Section 19.3.

While the reverse small-world experiments do not directly tell us about the structure of social networks, they do give us information about how people perceive and deal with social networks.

## 3.7 SNOWBALL SAMPLING, CONTACT TRACING, AND RANDOM WALKS

Finally in this chapter on social networks we take a look at a class of networkbased techniques for sampling hidden populations.

Studies of some populations, such as drug users or illegal immigrants, present special problems to the investigator because the members of these populations do not usually want to be found and are often wary of giving interviews. Techniques have been developed, however, to sample these populations by making use of the social network that connects their members together. The most widely used such technique is *snowball sampling* [108, 127, 310].

Note that, unlike the other experimental techniques discussed in this chapter, snowball sampling is not intended as a technique for probing the structure of social networks. Rather, it is a technique for studying hidden populations that relies on social networks for its operation. It is important to keep this distinction clear. To judge by the literature, some professional social network analysts do not, and the results are often erroneous conclusions and bad science.

Standard techniques such as telephone surveys often do not work well when sampling hidden populations. An investigator calling a random telephone number and asking if anyone on the other end of the line uses drugs is unlikely to receive a useful answer. The target population in such cases is small, so the chances of finding one of its members by random search are also small, and when you do find one they will very likely be unwilling to discuss the highly personal and possibly illicit topic of the survey with an investigator they have never met before and have no reason to trust.

So investigators probe the population instead by getting some of its members to provide contact details for others. The typical survey starts off rather like a standard ego-centered network study (Section 3.2.1). You find one initial member of the population of interest and interview them about themselves. Then, upon gaining their confidence, you invite them also to name other members of the target population with whom they are acquainted. Then you go and find those acquaintances and interview *them* asking them also to name further contacts, and so forth through a succession of "waves" of sampling. Pretty soon the process "snowballs" and you have a large sample of your target population to work with.

Clearly this is a better way of finding a hidden population than random surveys, since each named individual is likely to be a member of the population, and you also have the advantage of an introduction to them from one of their acquaintances, which may make it more likely that they will talk to you. However, there are some serious problems with the method as well. In particular, snowball sampling gives highly biased samples. In the limit of a large number of waves, snowball sampling samples actors with probability proportional to their "eigenvector centrality" (see Section 7.2). Unfortunately, this limit is rarely reached in practice, and in any case the eigenvector centrality cannot be calculated without knowledge of the complete contact network, which by definition we don't have, making correction for the sampling bias difficult. In short, snowball sampling gives biased samples of populations and there is little we can do about it. Nonetheless, the technique is sufficiently useful for finding populations that are otherwise hard to pin down that it has been widely used, biases and all, in studies over the last few decades.

Sometimes, in the case of small target populations, a few waves of snowball sampling may find essentially all members of a local population, in which case the method can be regarded as returning data about the structure of the social network. If the contacts of each interviewed participant are recorded in the study, it should be possible to reconstruct the contact network when the study is complete. This has occasionally been done in such studies, although as noted above the object is more often to exploit the social network to find the population than to study the network itself.

A technique closely related to snowball sampling is *contact tracing*, which is essentially a form of snowball sampling applied to disease incidence. Some diseases, such as tuberculosis and HIV, are considered sufficiently serious that, when someone is discovered to be carrying them, an effort must be made to track down all those who might also have been infected. Thus, in most Western countries, when a patient tests positive for HIV, for instance, he or she will be questioned about recent sexual contacts, and possibly about other types of potentially disease-carrying contacts, such as needle sharing if the patient is an injection drug user. Then health authorities will make an effort to track down those contacts and test them also for HIV. The process is repeated with any who test positive, tracing their contacts as well, and so forth, until all leads have been exhausted. While the primary purpose of contract tracing is to curtail disease outbreaks and safeguard the health of the population, the process also produces data about the community through which a disease is spreading and such data have sometimes been used in scientific studies, particularly of sexually transmitted diseases, for which data may otherwise be hard to come by. Population samples derived from contact tracing studies display biases similar in type and magnitude to those seen in snowball sampling and should be treated with the same caution. Indeed, they contain extra biases as well, since contacts are rarely pursued when an individual tests negative for the disease in question, so the sample is necessarily dominated by carriers of the disease, who are themselves usually a biased sample of the population at large. Also, as with snowball sampling, contact tracing data can provide us with an experimental window on the structure of the contact network itself, but again we expect the data to be strongly biased, except in cases of small target populations for which the sampling process saturates.

There is another variant of snowball sampling that deals to some extent with the problems of bias in the sample. This is *random-walk sampling* [182, 310]. In this method one again starts with a single member of the target community and interviews them and determines their contacts. Then, however, instead of interviewing all of those contacts, one chooses one of them at random and interviews only that one at the next step. If the person in question cannot be found or declines to be interviewed, one simply chooses another contact, and the process is repeated. Initially it appears that this will be a more laborious process than standard snowball sampling, since one spends a lot of time determining the names of individuals one never interviews, but this is not the case. In either method one has to determine the contacts of each person interviewed, so the total amount of work for a sample of a given size is the same. It is however very important that one really does determine all the contacts of each individual, even though most of the time only one of them is pursued. This is because for the method to work correctly one must make a random choice among those contacts, for example by rolling a die (or some modern electronic version thereof). To do this one must know the full set of contacts one is choosing between.

The advantage of the random-walk sampling method is that, as shown in Section 6.14, the asymptotic sampling probability of vertices in a random walk is simply proportional to vertex degree (see Eq. (6.60)). What's more, the asymptotic regime in such studies is, unlike snowball sampling, reached quite quickly for relatively small sample sizes. [26]

Knowing this, and given that we determine degree (i.e., the number of contacts an individual has) as a part of the interview process, we can easily compensate for sampling bias and make population estimates of quantities in a way that is, in theory at least, unbiased. In practice, many sources of bias remain, particularly those associated with participant subjectivity, inability to recall contacts, and non-participation of named contacts. Still, random-walk sampling is a great improvement on standard snowball sampling, and should be used more than it is. Its principal disadvantage is that it is relatively slow. Since the participants are interviewed serially, in a chain, rather than in parallel waves, a strict implementation of the method can take a long time to develop a large sample. One can get around this obstacle to some extent by running several short random walks in parallel instead of one long one, but the walks cannot be too short or they will not reach the asymptotic regime in which sampling is proportional to degree.

Another variant of the random-walk sampling idea is used to deal with a different problem, that of enrolling study participants. In some cases it is considered unethical to get participants to name their contacts, particularly when the topic of the study is one of dubious legality, and permission to perform such studies may be withheld by the authorities. To circumvent this problem one can make use of *respondent-driven sampling* [289]. In this technique, participants are usually paid to

take part, and enrollment is achieved by handing out tickets to interviewees. Rather than asking people to name their contacts, the interviewees are simply told that they should give the tickets to their friends, and that both they and the friends will receive payment if the friend brings the ticket to the investigator and agrees to participate in the survey. In this way, no one is ever asked to name names and all participants have actively volunteered their participation. In the case where a single ticket is given to each participant, the method is roughly equivalent to random-walk sampling and should in theory give a less biased sample than snowball sampling for the same reasons. In practice, a new bias is introduced because the recipient of the ticket is not necessarily chosen at random from an individual's acquaintances. Also, tickets frequently get lost or their recipients decline to participate, remuneration notwithstanding, so one would normally give out more than one ticket to each participant, which complicates the sampling process. Even so, it is believed that respondent-driven sampling provides superior population samples to snowball sampling, and it is the method of choice for studies in which one cannot ask people to name their contacts.